FORENSICS IN DIGITAL SECURITY (FDS)

Digital Forensic Tools I

OBJECTIVES

- Explain how to evaluate needs for computer forensics tools
- Describe available computer forensics software tools
- List some considerations for computer forensics hardware tools
- Describe methods for validating and testing computer forensics tools

COMPUTER FORENSICS TOOL NEEDS

Versatility

Flexibility

Robustness

+

OS

File system

Script capabilities

Automated features

Vendor's reputation

Application files

= ?

NEED

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

TYPES OF COMPUTER FORENSICS TOOLS

Hardware forensic tools	Software forensic tools

• Range from single-purpose components to complete computer systems and servers	• Commonly used to copy data from a suspect's disk drive to an image file
	• Types
	  • Command-line
	  • GUI

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK CATEGORIES

Acquisition	• Making a copy of the original drive
Validation and Discrimination	• Ensuring data integrity and sorting of data
Extraction	• Recovering data from evidence
Reconstruction	• Re-establish sequence of events of incident
Reporting	• Complete forensic analysis and prepare report

FORENSICS TASK

(ACQUISITION)

- Physical data copy
- Logical data copy
- Data acquisition format
- Command-line acquisition
- GUI acquisition
- Remote acquisition
- Verification

FORENSICS TASK
(ACQUISITION)

Data Acquisition

- Physical copying of the entire drive
- Logical copying of a disk partition

- All forensics acquisition tools can verify data-copying process.
- Vendor acquisition tools typically create smaller segmented files.

Data Acquisition Formats

Varies from generic to specific

Raw Data → Vendor-specific proprietary compressed format

View By Any Hexadecimal Editor

View By Vendor Software

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK

(ACQUISITION)

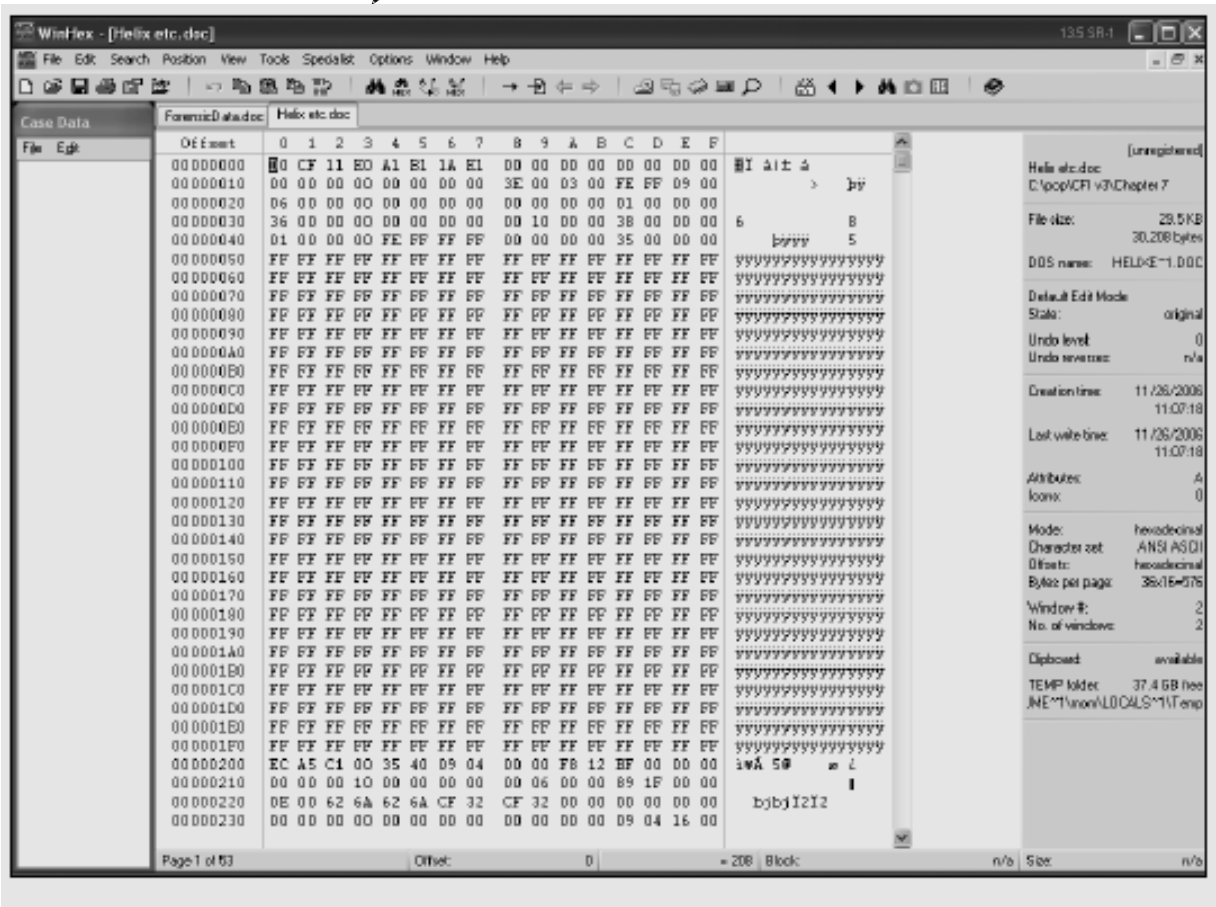


Figure 7.1 Sample Hexadecimal Editor Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK

(VALIDATION AND DISCRIMINATION)

Validation

- Ensure integrity of copied data

Discrimination

- Involves sorting and searching through all investigation data

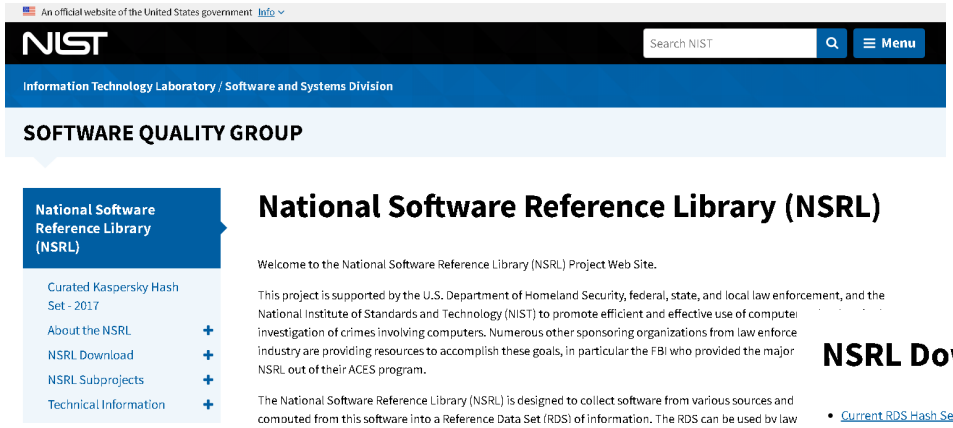
Key Functions

Hashing	Filtering	Analyzing file headers
• CRC-32, MD5, Secure Hash Algorithms	• Based on hash value sets	• Discriminate files based on their types

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK

(VALIDATION AND DISCRIMINATION)



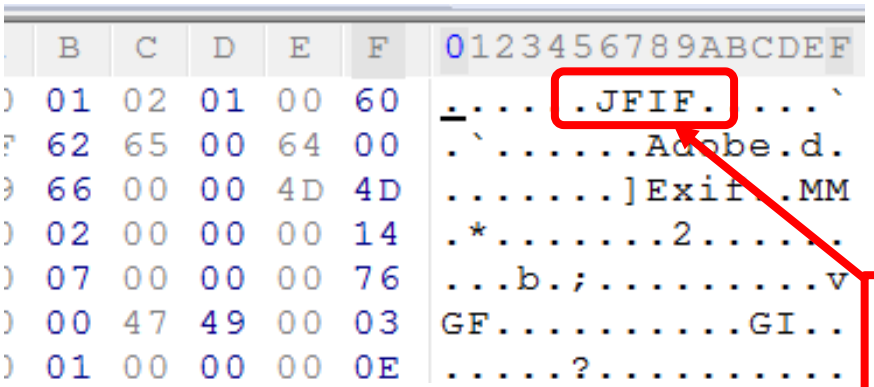
https://www.nist.gov/itl/ssd/software-quality-group/national-software-reference-library-nsrl/nsrl-download

NSRL Download

- Current RDS Hash Set
- Non-RDS Hash Sets
- RDS Query Tools
- Legacy Tools

National Software Reference Library (NSRL)

- Compiled a list of known file hashes
- For a variety of OSs, applications, and images



Most forensics tools

- Include a list of common header values
- Identify header values

Check file type for incorrect file extension

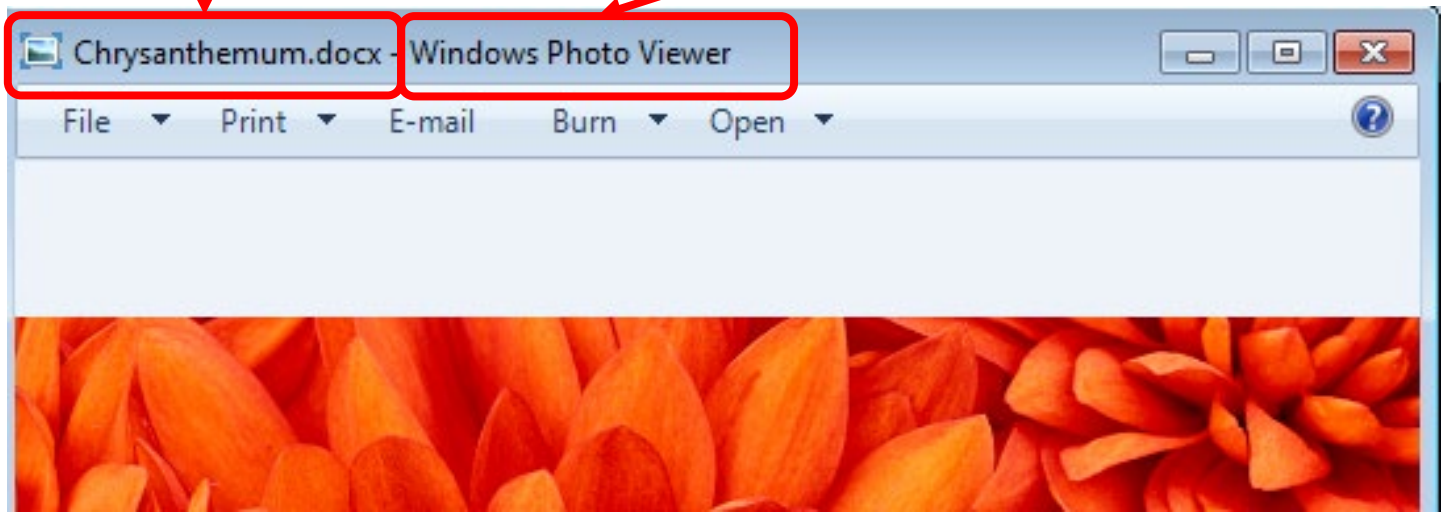
FORENSICS TASK

(VALIDATION AND DISCRIMINATION)

Examples to identify incorrect extension

Extension docx File

File is opened in Windows Photo Viewer



FORENSICS TASK
(EXTRACTION)

Recovering data is the most demanding first step in analyzing an investigation's data

Key tasks may include

Keywords search speeds up analysis

- Data viewing
- Keyword searching
- Decompressing
- Carving
- Decrypting
- Bookmarking

FORENSICS TASK
(EXTRACTION)

A Key Challenge

Encrypted files and systems → **A Problem**

Performs Password Dictionary Attack

If all fails

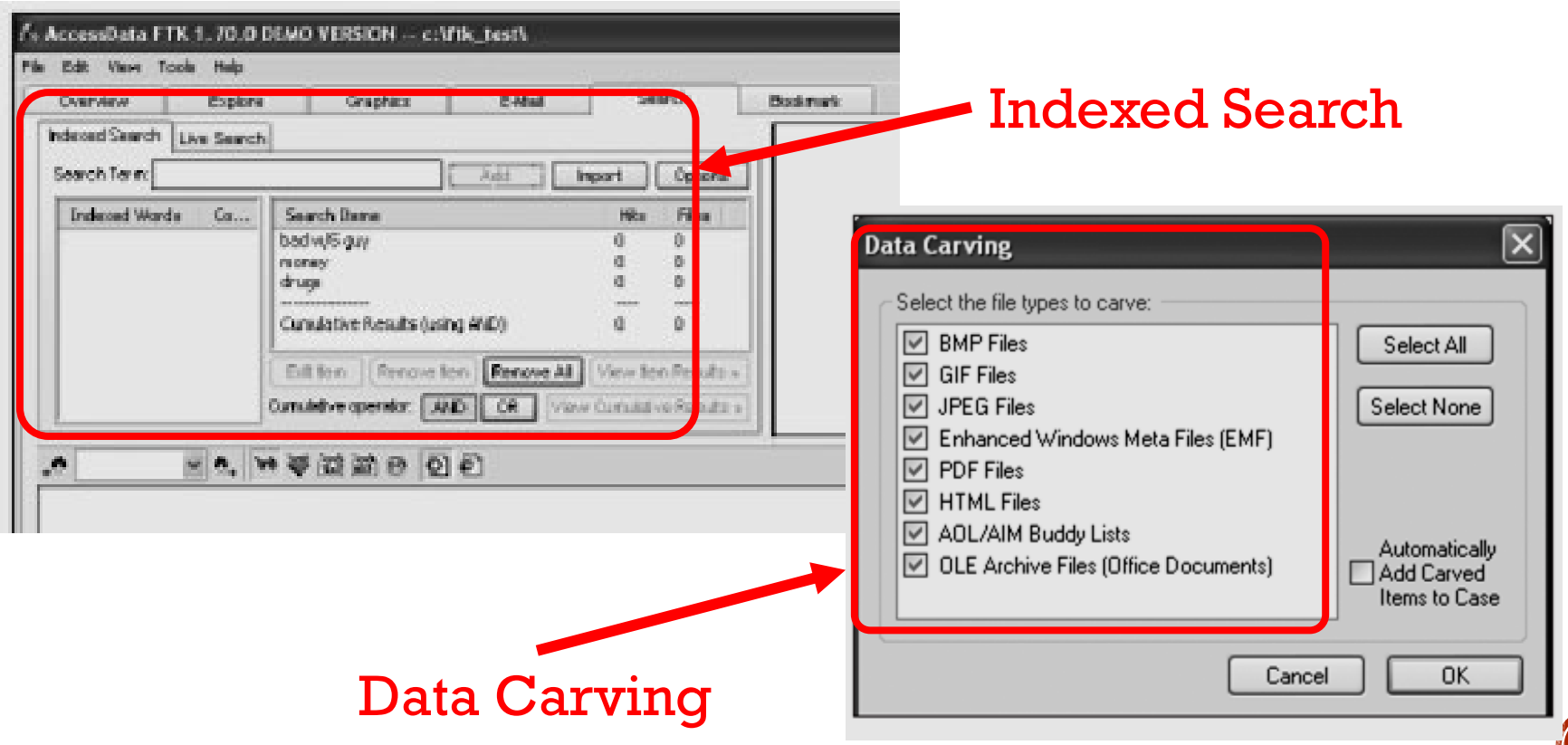
Brute-force attack

Use password recovery tools to recover password

FORENSICS TASK

(EXTRACTION)

Features in FTK



Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK
(RECONSTRUCTION)

Re-create a suspect drive to show what happened during a crime or an incident

Approaches

- Disk-to-disk copy
- Image-to-disk copy
- Partition-to-partition copy
- Image-to-partition copy

Some Imaging Tools

- SafeBack
- SnapBack
- EnCase
- FTK Imager
- ProDiscover

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TASK

(REPORTING)

Complete a forensics disk analysis and examination

Use findings to create an investigation report

Key Features

Log reports

Report generator

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

(COMPARISON)

Table 7-1 Comparison of forensics tool functions

Function	ProDiscover Basic	AccessData Ultimate Toolkit	Guidance Software EnCase
Acquisition			
Physical data copy	√	√	√
Logical data copy	√	√	√
Data acquisition formats	√	√	√
Command-line process			√
GUI process	√	√	√
Remote acquisition			√*
Verification	√	√	√
Validation and discrimination			
Hashing	√	√**	√**
Filtering		√	√
Analyzing file headers		√	√
Extraction			
Data viewing	√	√***	√***
Keyword searching	√	√	√
Decompressing		√	√
Carving		√	√
Decrypting		√	
Bookmarking	√	√	√
Reconstruction			
Disk-to-disk copy	√	√	√
Image-to-disk copy	√	√	√
Partition-to-partition copy	√		√
Image-to-partition copy	√		√
Reporting			
Log reports		√	√
Report generator	√	√	

Table 7-1 Comparison of forensics tools. Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

A Comparative Study based Digital Forensic Tool: Complete Automated Tool - Comparison of tools for each forensic phase

A Comparative Study based Digital Forensic Tool: Complete Automated Tool 28

Phases of proposed framework		Digital Forensic Tools																							
Sr. No	Phases	1 Encase[4][5][6]	2 FTK[7][8]	3 SANS SIFT [9]	4 PTK [10]	5 Bulk Extractor [11]	6 The Sleuth Kit[12]	7 The Coroner's [13]	8 DFF[14]	9 COFEE [15]	10 ProDiscover Basic [16]	11 Volatility[17]	12 Linux'dd'[18]	13 CAINE [19]	14 Recuva[20]	15 HexEditor [21]	16 DEFT[22]	17 Xplico [23]	18 LastActivityView[24]	19 [25]	20 PlanSight [26]	21 HxD[27]	22 HELIX3[28]	23 P2explorer [29]	24 [30]	25 USBDeview [31]
1	Acquire																		√							
2	Analyse	√	√								√	√				√				√	√		√		√	√
3	Approach Strategy																									
4	Assess																									
5	Attribute																						√			
6	Authenticate	√	√	√	√	√	√	√	√	√																
30	Readiness																									
31	Reconstruct	√	√																							
32	Recover	√	√	√	√	√	√	√	√	√				√	√		√									
33	Reduce	√	√														√									
34	Search										√				√	√		√				√				
35	Seizure																									
36	Submit																									
37	Trace															√			√							
38	Trace back																									
39	Transport																									
40	Triggering																									

Jain, N., & Kalbande, D. R. (2014). A comparative study based digital forensic tool: complete automated tool. *The International Journal of Forensic Computer Science*. http://ijofcs.org/V09N1-PP03-A-Comparative-Study.pdf

FORENSICS TOOLS
(SUMMARY)

Considerations

- Flexibility
- Reliability
- Expandability

Retains a software library containing older versions of forensics utilities, OSs, and other programs

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS
(COMMAND-LINE)

First Tools

- MS-DOS tools for IBM PC file systems, e.g. Norton DiskEdit
- Analyse floppy disks and hard disks

Advantages

- Command-line tools require few system resources
- Designed to run in minimal configurations

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

(UNIX/LINUX)

*nix Platforms

- *nix platforms have long been the primary command-line OSs

Helix

- One of the easiest suites
- Load in a live Windows system
- Loads as a bootable Linux OS from a cold boot

Autopsy and SleuthKit

- Sleuth Kit is a Linux forensics tool
- Autopsy is the GUI/browser interface used to access Sleuth Kit's tools

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

(UNIX/LINUX)

Knoppix-STD

- Linux bootable CD
- Collection of tools for configuring security measures, including computer and network forensics
- Forensically sound
  - Doesn't alter or damage the system you're analyzing

SMART

- Designed to be installed on numerous Linux versions
- Supports a variety of file systems
- Includes many plug-in utilities
- Hex viewer is available
- Commercial Product

More Tools

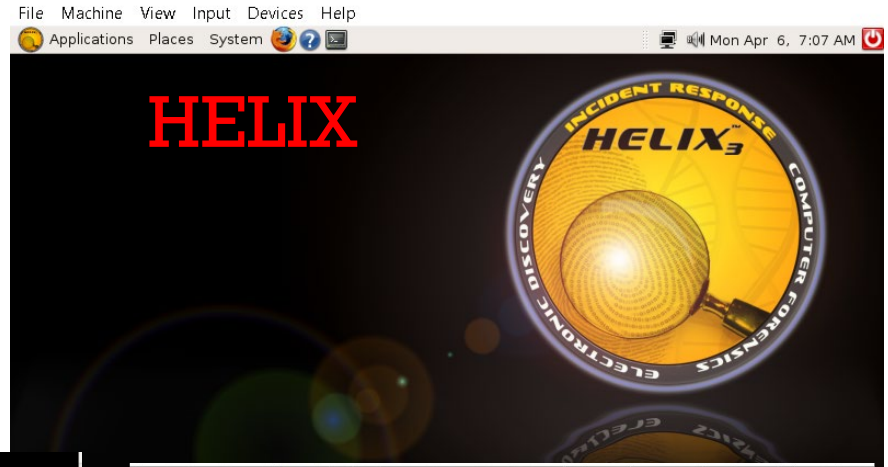
- SANS SIFT, DEFT, CAINE, Xplico

https://techtalk.gfi.com/top-20-free-digital-forensic-investigation-tools-for-sysadmins/

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

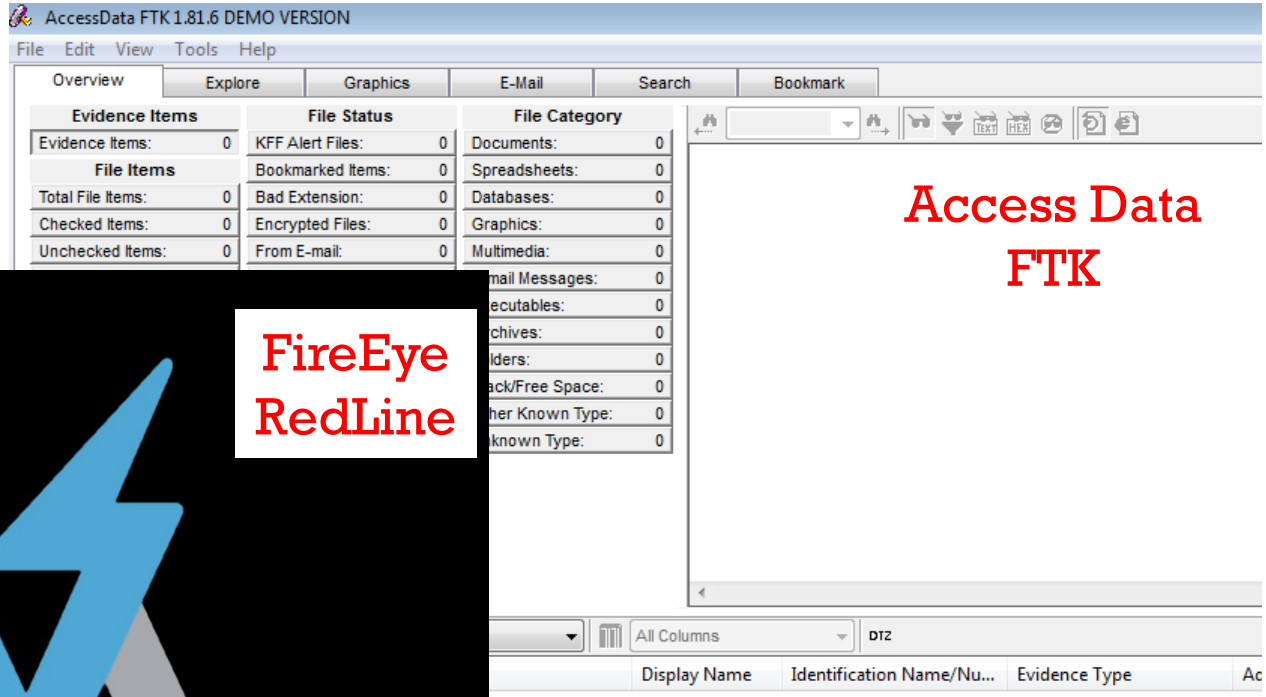
(UNIX/LINUX)



Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

(WINDOWS)



Access Data FTK

FireEye RedLine

FORENSICS TOOLS
(MORE GUI TOOLS)

- •Simplify computer forensics investigations
- Help training beginning investigators
- Most of them come into suites of tools

Advantages	Disadvantages
• Ease of use • Multitasking • No need for learning older Oss	• Excessive resource requirements • Produce inconsistent results • Create tool dependencies

Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.

FORENSICS TOOLS

(LINUX ACQUISITION)

Hash Functions

- Use for validation of acquired images
- e.g. md5sum, sha1sum, sha256sum, etc.

Example : Image from device (/dev/fd0)

Compute hash of device

e.g. `md5sum /dev/fd0`

Acquire image of device

`dd if='/dev/fd0' of=image_file`

Compute hash of acquired image

e.g. `md5sum image_file`

Outputs hash of device

Validate that hash value should match

Outputs hash of image

image_file – **used to search for evidence**

dc

```
dd if=/my/old/mbr of=/dev/sda bs=446 count=1
```

FORENSICS TOOLS

(LINUX ACQUISITION)

dd – Tool for image acquisition

if=<file>	• reads from file instead of STDIN
of=<file>	• write to file
ibs=<bytes>	• specify bytes per read
obs=<bytes>	• specify bytes per write
skip=<blocks>	• number of blocks to skip before copying starts
count=<blocks>	• number of blocks to copy

e.g. *dd if=/dev/sda bs=512 count=1 of=mbr.img*

FORENSICS TOOLS
(LINUX ACQUISITION)

What if the acquired image is large in size ?

e.g. large disk, images, RAID or backup tapes, etc

use ***split*** command to split images into smaller files

Compute hash of image file
e.g. `md5sum fileimg.img`

Outputs hash of device

Split image into smaller files
`e.g. split -b 1024M fileimg.img split_img_`

Combine split files
`e.g. cat split_img_* > rfileimg.img`

Validate that hash value should match

Compute hash of acquired image
e.g. `md5sum rfileimg.img`

Outputs hash of image

FORENSICS TOOLS

(USING DD FOR ACQUISITION IN WINDOWS)

dd for Windows @ http://www.chrysocome.net/dd

Examples

Make image from disk

- *dd if=\\.\a: of=c:\temp\disk1.img bs=1440k*

Make a disk from an image

- *dd if=c:\temp\disk1.img of=\\.\a: bs=1440k*

Extract .iso from a CD

- *dd if=\\?\Device\CdRom0 of=c:\temp\disc1.iso bs=1M*

Write to any file or block device in windows

Notation	Devices
\\.\	win32 exported devices
\\?\	windows native device

FORENSICS TOOLS

(WINDOWS ACQUISITION EXAMPLE)

Splitting image procedure with DD in windows

1. Hash the source

```
C:\bin>md5sum \\.\g:
\c226cdabd7d0423dcb8622f6171d6ded *\\\\.\\g:
```

2. Use dd to create image

```
C:\bin>dd if=\\.\g: of=c:\tmp\stick.img
rawwrite dd for windows version 0.4beta4.
Written by John Newbigin <jn@it.swin.edu.au>
This program is covered by the GPL.  See copying.txt for details
1007584+0 records in
1007584+0 records out
```

3. Take a baseline hash of image

```
C:\tmp>md5sum stick.img
c226cdabd7d0423dcb8622f6171d6ded *stick.img
```

FORENSICS TOOLS

(WINDOWS ACQUISITION EXAMPLE)

4. Create the image in parts

```
C:\tmp>split -b 200m stick.img stick.split.

C:\tmp>
C:\tmp>ls -lh
total 984M
-rwx------+   1 Malcolm  None        1.7k Aug 11 20:35 settings.txt
-rwx------+   1 Malcolm  None        492M Aug 23 23:11 stick.img
-rw-rw-rw-    1 Malcolm  None        200M Aug 23 23:23 stick.split.aa
-rw-rw-rw-    1 Malcolm  None        200M Aug 23 23:24 stick.split.ab
-rw-rw-rw-    1 Malcolm  None         92M Aug 23 23:24 stick.split.ac
drwx------+   2 Malcolm  None           0 Aug 14 23:30 winhex
```

5. Hash across the multiple parts

```
C:\tmp>cat stick.split.* | md5sum
c226cdabd7d0423dcb8622f6171d6ded *-
```

6. Restore the parts into one image

```
C:\tmp>cat stick.split.* > stick.new
```

7. Verify the new image

```
C:\tmp>md5sum stick.new
c226cdabd7d0423dcb8622f6171d6ded *stick.new
```

FORENSICS TOOLS

(VIEW FILES IN IMAGE)

How can you view the files in an image?

use ***mount*** command to open an image file as a file system

Example:

why is 'ro' important?

Mount image as file system

- `mount -o loop -o ro image_file directory_x`

View filestore of image file

- `ls directory_x`

- Different linux flavours have loopback and read only options

Mounted filestore can be viewed file by file and files can be searched

Allows examination of different filesystems e.g. MSDOS, traditional UNIX, RiscOS, MacOS, Amiga, CDROM, Windows NT filesystems

FORENSICS TOOLS
(NETWORK ACQUISITION)

How can you acquire an image over the network ?

use ***netcat*** to write data over network connection with TCP/IP

Combined with *dd* to provide a forensically sound method of obtaining image over the network

Necessary Equipment

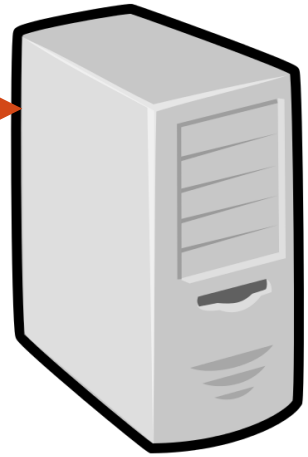
- Wiped hard drive for storage
- System to run *netcat* and mount storage
- Bootable forensic medias

FORENSICS TOOLS
(NETWORK ACQUISITION)

Target

This Photo by Unknown Author is licensed under CC BY-SA

Netcat Server



This Photo by Unknown Author is licensed under CC BY-SA

1. Setup server IP address
   `e.g. 192.168.0.1`
2. Start netcat server to listen on a particular port
   ```
   netcat -l -p 2000 -w 5 | pipebench > /mnt/test/image.dd
   ```
3. Access live target or boot from forensic tools
4. Setup target IP Address
   `e.g. 192.168.0.2`
5. Copy disk to server
   ```
   dd if=/dev/sdb conv=noerror,sync | pipebench | netcat 192.168.0.1 2000
   ```

FORENSICS TOOLS

(LINUX TOOLS)

xxd

- Provides a hex or ascii dump of a file (e.g., an image file) so it is searchable for hex or ascii sequences

grep

- Locates patterns in a file and outputs the line in which they are located

strings

- Prints out the readable characters from a file
- By default, print out strings from a file that are at least four characters long
- Useful for looking at data files without the originating program, and searching executables for useful strings, etc.

FORENSICS TOOLS

(DATA CARVING)

Data Carving

1. Locate boundaries of a file with known file signature
2. Pulling the data off of the drive within those boundaries

Carving JPEG

File type	Header bytes	Footer bytes
JPEG image file (jpg, jpeg)	0xffd8ffe	0xffd9

1. Use grep to identify start and end of jpeg

```
e.g. Start - xxd stick.img | grep "ffd8 " or
             xxd stick.img | grep "ff d8"
e.g. End   - xxd -s 123ABC stick.img | grep "ffd9 " or
             xxd -s 123ABC stick.img | grep "ff d9"
```

2. Use dd between start and end to extract image

```
e.g. dd if=stick.img of=carve.jpg skip=110264320 bs=1 count=2192
```

Messier, R. (2015). Operating System Forensics. Syngress.

FORENSICS TOOLS
(DATA CARVING)

Header bytes	Footer bytes	File type
0xffd8ffe	0xffd9	JPEG image file (jpg, jpeg)
0x1f9d		Compressed file (zip, z) using Lempel-Ziv-Welch algorithm
0x1fa0		Compressed file (zip, z) using LZH algorithm
0x425a68		Bzip2 compressed file (bz2)
0x0000001466747970 71742020		QuickTime movie file (MOV)
0x89504e470d0a1a	Multiple chunks in each file. Each chunk ends with IEND	Portable network graphics (PNG) file
0x424d		BMP file
0xfffb 0x494433		MPEG-1 Layer 3 (MP3) file without an ID3 tag MP3 file with an ID3 tag
0xd0cf11e0		Microsoft Word document (doc)
0x664c6143		Free Lossless Audio Codec (flac)
0x52494646		Waveform audio (wav)
0x4344303031		ISO9660 CD-ROM image file (iso)
0x25504446		Portable Document Format (PDF)
0x474946383761	0x003b	GIF

Adapted from Table 3.1 A Sample of File Signatures Messier, R. (2015). Operating System Forensics. Syngress.

FORENSICS TOOLS
(ADDITIONAL TOOLS)

The Coroners Toolkit (TCT)

- A package of utilities written by Dan Farmer and Wietse Venema
- TCTUtils
  - Add-on to TCT
    - Written by Brian Carrier
    - Provide file name analysis and mapping between file system layers
  - No longer supported, replaced by Sleuthkit which has added support for FAT and NTFS

Autopsy Forensic Browser and Sleuthkit

- Acquisition
  - Analyses images generated via dd
  - File and image verification/authentication
- Analysis
  - Filesystems: NTFS, FAT, FFS, ext2fs, ext3fs

TCT - http://www.porcupine.org/forensics/tct.html
TSK - http://www.sleuthkit.org

MORE

CONCLUSION

- Acquire suitable forensics tools for each forensic phases
- Maintain a software library on forensic investigations
- Computer Forensics Tools
  - Software/Hardware, Command-line/GUI
- GUI Tools don't require the same level of computing expertise as command-line tools
- Always test your forensics tools

REFERENCES

- Brooks, C. L. (September 2014). *CHFI Computer Hacking Forensic Investigator Certification All-in-One Exam Guide.* McGraw-Hill.
- Kleiman, D. (August 2011). The Official CHFI Study Guide (Exam 312-49). Syngress.
- Nelson, B., Phillips, A., & Christopher, S. (2010). *Guide to Computer Forensics and Investigations, 4th Edition.* Cengage Learning.
- Carrier, B. (2005). *File System Forensic Analysis.* Addison-Wesley Professional.
- Messier, R. (2015). Operating System Forensics. Syngress.

EXTRA SLIDES

THE CORONERS TOOLKIT (TCT)

- grave-robber
  - A data capturing tool, most commonly used to gather i-node information for use by mactime which is a program for time-ordering files/directories in the file system based on modification, access, and change ("MAC" for short) i-node time stamps
- ils and ils2mac
  - opens the named device and lists inode information
  - by default, ils lists the inodes of removed files
  - used to provide information re deleted files including timeline information
- unrm and lazarus
  - Tools for recovery of deleted file space and for processing this (or other, e.g., RAM or swap space) memory to attempt to identify and recover lost or hidden data
- icat
  - Displays contents of a file identified by inode (rather than filename, as in cat)
  - can be used to display contents of files identified by unallocated inodes (if block pointers still valid)

BACK

AUTOPSY FORENSIC BROWSER AND SLEUTHKIT

- by Brian Carrier
  - http://www.sleuthkit.org
- Acquisition
  - Analyses images generated via dd
  - file and image verification/authentication via md5
- Analysis
  - Filesystems: NTFS, FAT, FFS, ext2fs, ext3fs
  - Timelining – comprehensive: file MAC times, textual
  - Case management/documentation – via both built-in reporting feature and browser features; per host timezone and skew settings
  - Known file filter – supports use of NIST NSRL hash database to ignore known files, and a custom 'ignore' list of files

(TASK = 'The @stake Sleuth Kit', the set of open source forensic analysis tools accessed via command line which are invoked by the Autopsy front-end)

BACK